Post Optimality or Sensitivity Analysis

1. Type I: Change in the objective function coefficients:

Ex: Given an LPP

$$Max\ Z = 3x_1 + 5x_2$$

Subject to: $x_1 \le 4,\ x_2 \le 6, 3x_1 + 2x_2 \le 18, x_1 \ge 0, x_2 \ge 0.$

The optimal simplex table is given by:

C_B	B. V.	b_i	x_1	x_2	s_1	s_2	s_3
0	s_1	2	0	0	1	$2/3$	$-1/3$
5	x_2	6	0	1	0	1	0
3	x_1	2	1	0	0	$-2/3$	$1/3$
$Z_j - C_j \rightarrow$		$Z = 36$	0	0	0	3	1

Use sensitivity analysis

1. Discuss the effect on the optimality of solution if the objective function is changed to $3x_1 + 3x_2$.
2. Discuss the effect on the optimality of solution if the objective function is changed to $3x_1 + x_2$.

Solution:

1. With changed objective function $3x_1 + 3x_2$, we need to calculate the net evaluations $(Z_j - C_j)$ rows:

		$C_j \rightarrow$	3	3	0	0	0
C_B	B. V.	b_i	x_1	x_2	s_1	s_2	s_3
0	s_1	2	0	0	1	$2/3$	$-1/3$
3	x_2	6	0	1	0	1	0
3	x_1	2	1	0	0	$-2/3$	$1/3$
$Z_j - C_j \rightarrow$		$Z = 36$	0	0	0	1	1

As $Z_j - C_j \geq 0 \; \forall \, j$, the optimality condition still hold. Therefore the optimal condition $x_1 = 2, x_2 = 6, Z_{max} = 36$ will be unchanged.

2.

		$C_j \rightarrow$	3	1	0	0	0	Min Ratio
C_B	B. V.	b_i	x_1	x_2	s_1	s_2	s_3	
0	s_1	2	0	0	1	2/3	$-1/3$	$3 \longrightarrow$
1	x_2	6	0	1	0	1	0	6
3	x_1	2	1	0	0	$-2/3$	1/3	-
$Z_j - C_j \rightarrow$			0	0	0	$-1 \uparrow$	1	
0	s_2	3	0	0	3/2	1	$-1/2$	
1	x_2	3	0	1	$-3/2$	0	1/2	
3	x_1	4	1	0	1	0	0	
$Z_j - C_j \rightarrow$			0	0	3/2	0	1/2	

As $Z_j - C_j \geq 0\ \forall\ j$, the optimality condition hold. Therefore the optimal condition will be changed to $x_1 = 4, x_2 = 3, Z_{max} = 15$.

<u>Ex:</u> Given an LPP

$$\boldsymbol{Max\ Z = 4x_1 + 6x_2 + 2x_3}$$

Subject to: $\boldsymbol{x_1 + x_2 + x_3 \le 3,\ x_1 + 4x_2 + 7x_3 \le 9, x_1 \ge 0, x_2 \ge 0, x_3 \ge 0.}$

The optimal simplex table is given by:

C_B	B. V.	b_i	x_1	x_2	x_3	s_1	s_2
4	x_1	1	1	0	-1	$4/3$	$-1/3$
6	x_2	2	0	1	2	$-1/3$	$1/3$
$Z_j - C_j \rightarrow$		$Z = 16$	0	0	6	$10/3$	$2/3$

Use sensitivity analysis to find

1. **The range of cost coefficient, in objective function of x_1 so that the current solution remains optimal.**
2. **The range of cost coefficient, in objective function of x_3 so that the current solution remains optimal.**

Solution:

1. Let C_1 be the cost coefficient of the variable x_1, and corresponding to this cost coefficient we need to calculate the net evaluation $Z_j - C_j$.

		$C_j \rightarrow$	C_1	6	2	0	0
C_B	B. V.	b_i	x_1	x_2	x_3	s_1	s_2
C_1	x_1	1	1	0	-1	$4/3$	$-1/3$
6	x_2	2	0	1	2	$-1/3$	$1/3$
$Z_j - C_j \rightarrow$		$Z = 36$	0	0	$-C_1 + 10$	$\frac{4C_1}{3} - 2$	$-\frac{C_1}{3} + 2$

If the above table will be a optimal simplex table, then

$$-C_1 + 10 \geq 0 \implies C_1 \leq 10,$$

$$\frac{4C_1}{3} - 2 \geq 0 \implies C_1 \geq 3/2,$$

$$-\frac{C_1}{3} + 2 \geq 0 \implies C_1 \leq 6.$$

Combining the above three condition, we get: $\frac{3}{2} \leq C_1 \leq 6$ (the range of the cost coefficient of x_1).

Solution:

2. Let C_3 be the cost coefficient of the variable x_3, and corresponding to this cost coefficient we need to calculate the net evaluation $Z_j - C_j$.

		$C_j \rightarrow$	4	6	C_3	0	0
C_B	B. V.	b_i	x_1	x_2	x_3	s_1	s_2
4	x_1	1	1	0	-1	$4/3$	$-1/3$
6	x_2	2	0	1	2	$-1/3$	$1/3$
$Z_j - C_j \rightarrow$		$Z = 36$	0	0	$-C_3 + 8$	$\frac{10}{3}$	$\frac{2}{3}$

If the above table will be a optimal simplex table, then

$$\boldsymbol{-C_3 + 8 \geq 0 \quad \Longrightarrow \quad C_3 \leq 8.}$$

2. Type II: Change in right-hand side in the constraints:

Ex: Given an LPP

$$Max\ Z = 4x_1 + 6x_2 + 2x_3$$

Subject to: $x_1 + x_2 + x_3 \leq 3,\ x_1 + 4x_2 + 7x_3 \leq 9, x_1 \geq 0, x_2 \geq 0, x_3 \geq 0.$

The optimal simplex table is given by:

C_B	B. V.	b_i	x_1	x_2	x_3	s_1	s_2
4	x_1	1	1	0	-1	$4/3$	$-1/3$
6	x_2	2	0	1	2	$-1/3$	$1/3$
$Z_j - C_j \rightarrow$		$Z = 16$	0	0	6	$10/3$	$2/3$

Use sensitivity analysis to discuss

1. **The effect of discrete change in the availability of resource from $(3, 9)^T$ to $(3, 10)^T$.**
2. **The effect of discrete change in the availability of resource from $(3, 9)^T$ to $(9, 3)^T$.**
3. **Which source should be increased or decreased to get best marginal increase in the values of objective function.**

Solution:

1. Since $\widehat{x_B} = B^{-1}\hat{b} = \begin{bmatrix} 4/3 & -1/3 \\ -1/3 & 1/3 \end{bmatrix}\begin{bmatrix} 3 \\ 10 \end{bmatrix} = \begin{bmatrix} 1/3 \\ 7/3 \end{bmatrix} \geq 0$, the current solution will be still feasible after changing the right hand side vector. The optimal value will be $Z = \begin{bmatrix} 4 & 6 \end{bmatrix}\begin{bmatrix} 1/3 \\ 7/3 \end{bmatrix} = 50/3$.

2. Since $\widehat{x_B} = B^{-1}\hat{b} = \begin{bmatrix} 4/3 & -1/3 \\ -1/3 & 1/3 \end{bmatrix}\begin{bmatrix} 9 \\ 3 \end{bmatrix} = \begin{bmatrix} 11 \\ -2 \end{bmatrix}$, so the solution is infeasible but the from the optimal table we can observe that the optimality condition achieved. So, in this stage we will apply the dual simplex method:

		$C_j \rightarrow$	4	6	2	0	0
C_B	B. V.	$b_i \downarrow$	x_1	x_2	x_3	s_1	s_2
4	x_1	11	1	0	-1	$4/3$	$-1/3$
6	x_2	$\mathbf{-2} \rightarrow$	0	1	2	$-1/3$	$1/3$
$Z_j - C_j \rightarrow$		$Z = 0$	0	0	6	$\frac{10}{3} \uparrow$	$\frac{2}{3}$
$\theta_{ij} \longrightarrow$			$-$	$-$	$-$	$\left\lvert\frac{10/3}{-1/3}\right\rvert$	$-$
4	x_1	3	1	4	7	0	1
0	s_1	6	0	-3	-6	1	-1
$Z_j - C_j \rightarrow$			0	10	26	0	4

Since $b_i \geq 0 \; \forall \, i$, the optimality achieved in the dual simplex table.

The optimal solution will be:
$x_1 = 3, x_2 = 0, \; Z = 12$.

3. Let the RHS of the first constraint is b_1. Then

$$\widehat{x_B} = B^{-1}\hat{b} = \begin{bmatrix} 4/3 & -1/3 \\ -1/3 & 1/3 \end{bmatrix}\begin{bmatrix} b_1 \\ 9 \end{bmatrix} = \begin{bmatrix} \frac{4b_1}{3} - 3 \\ -\frac{b_1}{3} + 3 \end{bmatrix}$$

. To make this solution feasible, we need

$$\frac{4b_1}{3} - 3 \geq 0 \quad \Rightarrow \quad b_1 \geq 9/4$$

$$-\frac{b_1}{3} + 3 \geq 0 \quad \Rightarrow \quad b_1 \leq 9$$

Combining both the restriction we get the range $\frac{9}{4} \leq b_1 \leq 9$.

Similarly, consider the RHS of second constraint is b_2, Then

$$\widehat{x_B} = B^{-1}\hat{b} = \begin{bmatrix} 4/3 & -1/3 \\ -1/3 & 1/3 \end{bmatrix}\begin{bmatrix} 3 \\ b_2 \end{bmatrix} = \begin{bmatrix} 4 - \frac{b_2}{3} \\ \frac{b_2}{3} - 1 \end{bmatrix}$$

. To make this solution feasible, we need

$$4 - \frac{b_2}{3} \geq 0 \quad \Rightarrow \quad b_2 \leq 12$$

$$\frac{b_2}{3} - 1 \geq 0 \quad \Rightarrow \quad b_2 \geq 3$$

Combining both the restriction we get the range $3 \leq b_2 \leq 12$.

3. Type III: Addition of a constraint:

Ex: The optimal simplex table of the above LPP is given by:

		$C_j \rightarrow$	6	5	0	0
C_B	B. V.	$b_i \downarrow$	x_1	x_2	s_1	s_2
5	x_2	3	0	1	3	-1
6	x_1	2	1	0	-2	1
$Z_j - C_j \longrightarrow$			0	0	3	1

Use sensitivity analysis

1. **Discuss the effect on the optimality of solution if we add an additional constraint $2x_1 + x_2 \leq 7$.**
2. **Discuss the effect on the optimality of solution if if we add an additional constraint $2x_1 + x_2 \geq 8$.**

Solution:

1. The solution $x_1 = 2,\ x_2 = 3$ satisfies the constraint $\boldsymbol{2x_1 + x_2 \leq 7}$, so the optimal solution still unchanged after adding the new constraint.

1. But the solution $x_1 = 2,\ x_2 = 3$ does not satisfy the constraint $\boldsymbol{2x_1 + x_2 \geq 8}$, so the optimal solution will be affected by the addition of such constraint. The new solution can be obtained as follows:

$$\boldsymbol{2x_1 + x_2 \geq 8 \ \Rightarrow -2x_1 - x_2 \leq -8}$$

		$C_j \rightarrow$	6	5	0	0	0
C_B	B. V.	$b_i \downarrow$	x_1	x_2	s_1	s_2	s_3
5	x_2	3	0	1	3	− 1	0
6	x_1	2	1	0	− 2	1	0
0	s_3	−8	−2	− 1	0	0	1
5	x_2	3	0	1	3	− 1	0
6	x_1	2	1	0	− 2	1	0
0	s_3	−4	0	− 1	− 4	2	1
5	x_2	3	0	1	3	− 1	0
6	x_1	2	1	0	− 2	1	0
0	s_3	$\mathbf{-1} \longrightarrow$	0	0	− 1	1	1
$Z_j - C_j \longrightarrow$			0	0	3	1	0
$\theta_{ij} \longrightarrow$			–	–	$\left\lvert\frac{3}{-1}\right\rvert$	–	–
5	x_2	**0**	0	1	0	2	3
6	x_1	**4**	1	0	0	− 1	− 2
0	s_1	**1**	0	0	1	− 1	− 1
$Z_j - C_j \longrightarrow$			0	0	0	4	3

Since $Z_j - C_j \geq 0 \; \forall \, j, and \; b_i \geq 0 \;\; \forall \, i$, the optimality criterial holds in the dual simplex table.

The optimal solution is :
$\boldsymbol{x_1 = 4, x_2 = 0, Z_{max} = 24.}$

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<u>Ex:</u> **Suppose the optimal simplex table is given by:**

		$C_j \rightarrow$	2	1	-1	0	0
C_B	B. V.	$b_i \downarrow$	x_1	x_2	x_3	s_1	s_2
2	x_1	8	1	1	1	1	0
0	s_2	12	0	3	-1	1	1
$Z_j - C_j \rightarrow$			0	1	3	2	0

Use sensitivity analysis to determine the change in the optimal solution if we add the constraint $\boldsymbol{x_2 + x_3 \geq 3}$ in the final table.

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Solution:

		$C_j \rightarrow$	2	1	-1	0	0	0
C_B	B. V.	$b_i \downarrow$	x_1	x_2	x_3	s_1	s_2	s_3
2	x_1	8	1	1	1	1	0	0
0	s_2	12	0	3	-1	1	1	0
0	s_3	$-3 \longrightarrow$	0	-1	-1	0	0	1
$Z_j - C_j \rightarrow$			0	$1 \uparrow$	3	2	0	0
$\theta_{ij} \longrightarrow$			$-$	$\left\|\frac{1}{-1}\right\|$	$\left\|\frac{3}{-1}\right\|$	$-$	$-$	$-$
2	x_1	5	1	0	0	1	0	1
0	s_2	3	0	0	-4	1	1	3
1	x_2	3	0	1	1	0	0	-1
			0	0	1	2	0	1

Since $Z_j - C_j \geq 0\ \forall\ j$, and $b_i \geq 0\ \ \forall\ i$, the optimality criterial holds in the dual simplex table.

The optimal solution is :
$x_1 = 5, x_2 = 3,\ \ x_3 = 0,$
$Z_{max} = 13.$

4. Type IV: Addition of a decision variable:

Ex: **Consider the following LPP:**

$$\boldsymbol{Min\ Z = 2x_1 + x_2}$$

$$\textbf{Subject to: } \boldsymbol{3x_1 + x_2 = 3,}$$
$$\boldsymbol{4x_1 + 3x_2 \geq 6,}$$
$$\boldsymbol{x_1 + 2x_2 \leq 3, \quad x_1, x_2 \geq 0.}$$

The optimal simplex table of the above LPP is given by:

		$C_j \rightarrow$	2	1	0	M	M	0
C_B	B. V.	$b_i \downarrow$	x_1	x_2	s_2	A_1	A_2	s_3
2	x_1	$3/5$	1	0	$1/5$	$3/5$	$-1/5$	0
1	x_2	$6/5$	0	1	$-3/5$	$-4/5$	$3/5$	0
0	s_3	0	0	0	0	1	-1	1
$Z_j - C_j \rightarrow$		$Z = 27$	0	0	$-1/5$	$\frac{2}{5} - M$	$\frac{1}{5} - M$	0

What if we add a new activity $x_3 \geq 0$, with cost $1/2$ in the objective function and the constraint coefficient as $(0, 5, 2)^T$.

Ex: After adding the variable, the new LPP is given by:

$$Min\ Z = 2x_1 + x_2 + \frac{1}{2}x_3$$

$$\text{Subject to: } 3x_1 + x_2 + 0.x_3 = 3,$$
$$4x_1 + 3x_2 + 5.x_3 \geq 6,$$
$$x_1 + 2x_2 + 2.x_3 \leq 3, \quad x_1, x_2 \geq 0.$$

As A_1, A_2 and s_3 are the starting variables in the initial simplex table, from the given final simplex table, we get

$$B^{-1} = \begin{bmatrix} 3/5 & -1/5 & 0 \\ -4/5 & 3/5 & 0 \\ 1 & -1 & 1 \end{bmatrix}.$$

So, the corresponding column of x_3 in the final table: $\alpha_3 = B^{-1}\widehat{A_3} = \begin{bmatrix} 3/5 & -1/5 & 0 \\ -4/5 & 3/5 & 0 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 5 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \\ -3 \end{bmatrix}.$

		$C_j \rightarrow$	2	1	0	M	M	0	1/2	Min Ratio
C_B	B. V.	$b_i \downarrow$	x_1	x_2	s_2	A_1	A_2	s_3	x_3	
2	x_1	3/5	1	0	1/5	3/5	$-1/5$	0	-1	-
1	x_2	6/5	0	1	$-3/5$	$-4/5$	3/5	0	3	$6/5 \longrightarrow$
0	s_3	0	0	0	0	1	-1	1	-3	-
$Z_j - C_j \rightarrow$		$Z = 27$	0	0	$-1/5$	$\frac{2}{5} - M$	$\frac{1}{5} - M$	0	$1/2 \uparrow$	
2	x_1	1	1	1/3	0	1/3	0	0	0	
1/2	x_3	2/5	0	1/3	$-1/5$	$-4/15$	1/5	0	1	
0	s_3		0	1	2/5	1/5	$-2/5$	1	0	
$Z_j - C_j \rightarrow$			0	$-\frac{1}{6}$	$-\frac{1}{10}$	$\frac{8}{15} - M$	$\frac{1}{10} - M$	0	0	

Since $Z_j - C_j \leq 0 \ \forall \ j$, and the given problem is minimization type, so the optimality condition achieved.

The updated optimal solution is

$$\boldsymbol{x_1 = 1, \qquad x_2 = 0, \qquad x_3 = \frac{2}{5}, \qquad Z_{min} = \frac{11}{5}.}$$

Ex: **Consider the following LPP:**

$$\boldsymbol{Max\ Z = 6x_1 + 5x_2}$$

$$\textbf{Subject to: } \boldsymbol{x_1 + x_2 \le 5,}$$

$$\boldsymbol{3x_1 + 2x_2 \le 12, \quad x_1, x_2 \ge 0.}$$

The optimal simplex table of the above LPP is given by:

		$C_j \rightarrow$	6	5	0	0
C_B	B. V.	$b_i \downarrow$	x_1	x_2	s_1	s_2
5	x_2	3	0	1	3	-1
6	x_1	2	1	0	-2	1
$Z_j - C_j \longrightarrow$			0	0	3	1

we add a new activity $\boldsymbol{x_3 \ge 0}$, with constraint coefficient as $\boldsymbol{(2, 3)^T}$. What should be the cost coefficient of $\boldsymbol{x_3}$ in the objective function such that the optimal solution remains unchanged.

As s_1, s_2 are the starting variables in the initial simplex table, from the given final simplex table, we get

$$B^{-1} = \begin{bmatrix} 3 & -1 \\ -2 & 1 \end{bmatrix}$$

So, the corresponding column of x_3 in the final table: $\alpha_3 = B^{-1}\widehat{A_3} = \begin{bmatrix} 3 & -1 \\ -2 & 1 \end{bmatrix}\begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$.

		C_j →	6	5	0	0	C_3
C_B	B. V.	b_i ↓	x_1	x_2	s_1	s_2	x_3
5	x_2	3	0	1	3	− 1	3
6	x_1	2	1	0	− 2	1	− 1
$Z_j - C_j$ ⟶			0	0	3	1	$9 - C_3$

Now, if the current solution will be optimal solution then, $9 - C_3 \geq 0 \quad \Rightarrow \quad C_3 \leq 9$.

5. Type V: Change in the coefficient of the decision variable in constraint:

When the variable is a non-basic variable:

Ex: Consider the following LPP:

$$Max\ Z = 6x_1 + 8x_2$$

$$\text{Subject to: } x_1 \le 4,$$

$$3x_1 + 2x_2 \le 18, \quad x_1, x_2 \ge 0.$$

The optimal simplex table of the above LPP is given by:

		$C_j \rightarrow$	6	8	0	0
C_B	B. V.	$b_i \downarrow$	x_1	x_2	s_1	s_2
0	s_1	4	1	0	1	0
8	x_2	9	3/2	1	0	1/2
$Z_j - C_j \longrightarrow$			6	0	0	4

What if the coefficient of the non basic variable x_1 is changed from $[1,3]^T$ to $[1,1]^T$.

The corresponding column of x_1 in the final table: $\alpha_1 = B^{-1}\widehat{A_1} = \begin{bmatrix} 1 & 0 \\ 0 & 1/2 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1/2 \end{bmatrix}.$

		$C_j \to$	6	8	0	0	Min Ratio
C_B	B. V.	$b_i \downarrow$	x_1	x_2	s_1	s_2	
0	s_1	4	1	0	1	0	$4 \rightarrow$
8	x_2	9	1/2	1	0	1/2	18
$Z_j - C_j \longrightarrow$			$-2\uparrow$	0	0	4	
6	x_1	4	1	0	1	0	
8	x_2	7	0	1	$-1/2$	1/2	
$Z_j - C_j \longrightarrow$			0	0	2	4	

Since $Z_j - C_j \geq 0\ \forall\, j$, and the given problem is maximization type, so the optimality condition achieved.
The updated optimal solution is

$$x_1 = 4, \qquad x_2 = 7, \qquad Z_{max} = 80.$$

When the variable is a basic variable:

Ex: **Consider the following LPP:**

$$\boldsymbol{Max\ Z = 6x_1 + 5x_2}$$
$$\textbf{Subject to: } \boldsymbol{x_1 + x_2 \leq 5,}$$
$$\boldsymbol{3x_1 + 2x_2 \leq 12, \quad x_1, x_2 \geq 0.}$$

The optimal simplex table of the above LPP is given by:

		$C_j \rightarrow$	6	5	0	0
C_B	B. V.	$b_i \downarrow$	x_1	x_2	s_1	s_2
0	x_2	3	0	1	3	-1
8	x_1	2	1	0	-2	1
$Z_j - C_j \longrightarrow$			0	0	3	1

What if the coefficient of the basic variable $\boldsymbol{x_1}$ is changed from $\boldsymbol{[1, 3]^T}$ to $\boldsymbol{[1, 1]^T}$.

Before changing the coefficient of x_1,

the matrix $B = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$, and from the given table $B^{-1} = \begin{bmatrix} 3 & -1 \\ -2 & 1 \end{bmatrix}$.

Since, we have changed the coefficient of x_1 and x_1 be a basic variable in the final table so,

$$B_{new} = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}, \qquad \Rightarrow \quad B^{-1} = \frac{adj(A)}{|A|} = \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix}.$$

Therefore, $b_{new} = B_{new}^{-1} b = \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 5 \\ 12 \end{bmatrix} = \begin{bmatrix} 7 \\ -2 \end{bmatrix}$.

Since, b_{new} is not non-negative so will proceed with the dual simplex method in the next table.

		$C_j \rightarrow$	6	8	0	0	
C_B	B. V.	$b_i \downarrow$	x_1	x_2	s_1	s_2	
8	x_2	7	0	1	− 1	1	
6	x_1	$-2 \rightarrow$	1	0	2	− 1	
$Z_j - C_j \longrightarrow$			0	0	7	$-1 \uparrow$	
$\theta_{ij} \longrightarrow$			–	–	–	$\left\lvert\frac{-1}{-1}\right\rvert$	Min Ratio
8	x_2	5	1	1	1	0	$5 \rightarrow$
0	s_2	2	−1	0	− 2	1	-
$Z_j - C_j \longrightarrow$			$-1 \uparrow$	0	5	0	
6	x_1	5	1	1	1	0	
0	s_2	7	0	1	− 1	1	
			0	1	6	0	

Since $Z_j - C_j \geq 0 \; \forall \; j$ in the final table, and the given problem is maximization type, so the optimality condition achieved.

The updated optimal solution is

$$x_1 = 5, \qquad x_2 = 0, \qquad Z_{max} = 30.$$

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